

EXECUTIVE PROJECT REPORT (DS458)

Domain C: Customer & Marketing Analytics

Strategic Business Consultancy & Predictive Modeling

1. Project Objective & Data Scope

This consultancy project aims to transform raw retail datasets into a predictive engine for strategic advantage. To ensure high statistical significance while maintaining processing efficiency, a random sample of **50,000 records** was extracted. The primary goal is to analyze customer behavior and optimize sales forecasting through integrated engineering and advanced analytics.

2. Phase 1: Data Engineering & Preprocessing (Excel)

We applied a rigorous multi-step cleaning process to ensure 100% data integrity:

- **Text Standardization:** Used TRIM and PROPER functions to fix product descriptions.
- **Deduplication:** Eliminated redundant records to ensure accurate revenue metrics.
- **Null Handling:** Filtered out records without a CustomerID to maintain valid user-transaction links.
- **Feature Engineering:** Created the Status (Completed/Canceled) column and calculated Total Sales.
- **Temporal Extraction:** Isolated Year, Month, and Hour for deep-dive trend analysis.

Key Finding: Preliminary Pivot Table analysis revealed that the UK market contributes over **80% of total revenue**, identifying a significant geographic concentration risk.

3. Phase 2: Descriptive BI & Interactive Results (Power BI)

Dashboard Architecture

- **KPI Achievement:** Successfully tracked **\$1.05M** in total sales against a \$1.5M target.
- **Segmentation:** Dynamic slicers isolated "Priority" transactions (>\$100) from "Standard" retail activity.
- **Visual Insights:** Utilized Gauges, Treemaps, and Waterfall charts to identify primary cost contributors (Manual & Postage).

Advanced Data Modeling

Established a **Many-to-One (*:1)** relationship between sales and product tables. We implemented a dedicated Calendar Table to calculate Year-To-Date (YTD) metrics and Year-over-Year (YoY) growth, which peaked at **+47.26%** for December.

4. Phase 3: Predictive & Advanced Analytics (Python)

Revenue Forecasting

Using an AI-driven **Exponential Smoothing model**, the forecast projects a revenue plateau in the first half of 2012, signaling the need for immediate market intervention.

Statistical Anomaly Detection

Integrated **Python-based Z-Score filtering** to identify "Extreme Risk" transactions and entry errors, ensuring the reporting layer is free from skewed data.

XAI: Explainable AI (SHAP)

Utilized **SHAP Force Plots** to audit bulk transactions. This confirmed that massive outliers (e.g., a single order of 488k units) were legitimate UK-based wholesale orders rather than data entry mistakes.

5. Strategic Recommendations (Executive Actions)

Market Diversification: Aggressively expand marketing efforts in Saudi Arabia and emerging regions to reduce UK market dependency.

VIP Loyalty Program: Implement a tiered rewards system for "Priority" segment high-spenders to secure long-term retention.

Inventory Forecasting: Increase stock safety levels for top-performers (e.g., Regency Cakestand) at least 60 days prior to the November peak.

Data Quality Automation: Deploy real-time validation scripts to catch manual entry errors at the source of transaction.

6. Ethics, Privacy & Professionalism

Principle	Implementation Measure
Data Privacy	Customer identities were protected using anonymized IDs. All PII was excluded from the modeling layer.
Statistical Honesty	Used Z-Score auditing to ensure management decisions are based on accurate, normalized distributions.
Algorithmic Accountability	Manual SHAP auditing was performed to ensure AI-driven insights are logical and free from algorithmic bias.

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